

Marwan Badr

Software Developer | Backend, ERP & AI Systems

marwanbadr514@gmail.com · +20 100 725 8086
github.com/Marwan-badr543 · linkedin.com/in/marwan-badr-4553a42bb
portfolio · Cairo, Egypt



PROFESSIONAL SUMMARY

Software Developer with deep specialization in ERP engineering (**Odoo** and **Frappe Framework** / **ERPNext**), robust backend architectures, and autonomous **Agentic AI** systems. Combines a Bachelor of Commerce (English Accounting) background with production experience in Python, FastAPI, and LangGraph to architect high-performance APIs, custom business modules, real-time sync engines, and guarded LLM-driven agents. Proven track record building end-to-end multi-agent SaaS ecosystems, custom ERP addons, and transactional financial workflows with zero-compromise data integrity.

TECHNICAL SKILLS

Odoo Development	Odoo 16/17/18, Custom Addon Architecture, ORM, Models, XML Views (Tree, Form, Kanban), QWeb Reports, Wizards, Server Actions, Automated Actions, In-ERP Web Client UI (OWL/JS), Security Groups & Record Rules (<code>ir.rule</code> , <code>ir.model.access</code>), External APIs (XML-RPC, JSON-RPC), Accounting & Sales Workflows
Frappe & ERPNext	Frappe Framework v14/v15, ERPNext Customization, DocType Modeling, Server Scripts, Client Scripts, Whitelisted Python APIs, Hooks Architecture, Jinja Print Formats, Workflows & State Machines, Role Permissions Manager, Custom Frappe Apps, Background Jobs & Event Hooks
AI & Agentic Systems	LangGraph, LangChain, Tool Calling, Multi-Agent Architecture (Router & Desks), RAG, Structured Output, Qdrant Vector DB, Token Cost Metering, Prompt Engineering
Backend & API	Python, FastAPI, Django, Django REST Framework (DRF), RESTful APIs, Webhooks, Celery, Huey, AsyncIO
Databases & Caching Languages	PostgreSQL (Row-Level Security / RLS), MySQL, MariaDB, Redis (Caching & Task Queues)
DevOps & Tools	Python, JavaScript, HTML5, CSS3, SQL
Accounting Domain	Docker, Git, GitHub, Nginx (Zero-Copy X-Accel), Linux / Bash, Systemd
Desktop Development	Professional Financial Accounting (PFA), Chart of Accounts, General Ledger, AR/AP Reconciliations, 10-Tier Bank Reconciliation, Period Audits, Financial Statement Generation
Spoken Languages	PyQt5
	Arabic (Native), English (Very Good – English Commerce Section), French (Beginner)

PROFESSIONAL EXPERIENCE

Founder & Lead Architect

Aug 2026 – Present

Razyyn AI (Autonomous Accounting Virtual Finance Dept SaaS) – Remote

- **Complete Ecosystem Development:** Architected and built the entire technical ecosystem from scratch: multi-agent AI system (LangGraph orchestrator, Master Router, 6 specialized desks), high-throughput async backend (FastAPI), operator admin panel (React/Vite), multilingual landing platform (Next.js), native Frappe apps (v14/v15), and native Odoo modules (v17/v18).
- **Autonomous Multi-Agent Architecture:** Engineered the core agentic engine coordinating 6 isolated accounting desks (Ask, Analyse, Reconcile, Audit, Record, Update) with dynamic checklist planning, resumable state checkpoints, and structured output extraction.
- **Dual-Gated Write Safety:** Built two-layer transactional safety requiring ERP RBAC verification, Razyyn Agent Write Policy rules (monetary thresholds, blocked accounts, batch limits), and interactive human approval cards before committing ERP writes.
- **Financial Engineering & Algorithms:** Developed 10-tier algorithmic bank reconciliation resolving 100% of ledger-to-statement pairings, and AST SQL query rewrites (`sqlglot`) isolating live posted records while eliminating draft/cancelled entries.
- **Product Strategy & Marketing:** Led end-to-end product roadmap, financial data modeling, sales kits, and technical marketing across MENA.

Software / Frappe Developer

Mar 2026 – Present

Showsatellite (Home Appliances Retail) – Remote (Bahrain)

- Lead custom **Frappe Framework & ERPNext** engineering, DocType modeling, custom client/server scripts, print formats, and workflow automations for retail and enterprise operations.
- Engineered a high-throughput, real-time bidirectional synchronization engine connecting **ERPNext** and **OpenCart**, automating catalog updates, multi-tier pricing, stock levels, and product images with zero variance.
- Built a centralized operations logistics dashboard managing TV/AC delivery time-slot scheduling, real-time driver dispatching, and field technician ticketing.
- Developed and maintained integrated customer-facing e-commerce web applications with real-time sales order generation and Telegram alert webhooks.

- Managed daily accounting cycles, journal entries, Chart of Accounts maintenance, and Accounts Receivable/Payable (AR/AP) reconciliations.

FEATURED PROJECTS

Autonomous AI & Enterprise Ecosystem (Razyyn AI)

- **Razyyn AI — Autonomous AI Accountant Platform (Founder & Architect):** Multi-agent SaaS (FastAPI, LangGraph, Qdrant, Redis) for ERPNext and Odoo. Orchestrates a Master Router and 6 specialized accounting desks (Ask, Analyse, Reconcile, Audit, Record, Update), executing automated period audits, 10-tier bank reconciliation, and guarded transactional ERP writes.
- **Razyyn AI — Native Odoo Modules (v17 & v18):** Custom Odoo addons embedding in-ERP chat UI, guarded write gateway, AST-based live row SQL filtering, and multi-company security.
- **Razyyn AI — Native Frappe Applications (v14 & v15):** Custom Frappe apps providing ERPNext chat UI, multi-agent modes (Ask, Analyse, Audit), attachment processing, and granular write policy controls.

Web & Backend Architecture Projects

- **File Frog — High-Throughput Document Processing & Conversion Engine:** High-efficiency single-server file processing and conversion platform (Next.js, FastAPI, Redis, Huey, Nginx). Features chunked memory streaming (64KB chunks), system-wide RAM/disk backpressure controls, segmented worker queues, sandboxed LibreOffice CLI conversions, Tesseract OCR, and zero-copy Nginx X-Accel-Redirect downloads.
- **Integrated E-Commerce & Enterprise Operations Platform:** Full-scale customer ordering platform integrated with ERPNext, automating sales workflows, Telegram notifications, Google Sheets synchronization, and logistics dispatch management.
- **Real-Time Synchronization Engine: ERPNext ↔ OpenCart:** Production bidirectional sync engine automating product catalogs, prices, inventory, and high-speed image transfers between ERPNext and OpenCart with automated failover and activity logging.
- **Office Leaks — Professional Social Network Platform:** Corporate transparency social platform built with Django and Django REST Framework (DRF), featuring company workplace reviews, culture ratings, nested discussions, and secure JWT authentication.

Odoo ERP Custom Addons

- **Odoo Sales Commission Management Module:** Custom Odoo 17 module automating sales team commission tracking, computation rules linked directly to invoice records, and commission payout audit wizards.
- **Odoo Property Management Module:** Real estate property management addon for Odoo 17 enabling property listings, expected/selling pricing, owner profile linkages, and automated sales lifecycle transitions.
- **Odoo To-Do List & Timesheet Module:** Task management addon for Odoo 17 with task state workflows, validation rules preventing overtime allocation, and timesheet tracking.

Frappe Framework & ERPNext Custom Applications

- **Support Management Frappe App:** Comprehensive support ticketing and field service app for Frappe/ERPNext, managing maintenance appointment schedules, technician allocation, and customer requests.
- **Telegram Notification Frappe App:** Dynamic event-driven Frappe application that sends real-time Telegram alerts with interactive inline keyboard buttons (Submit/Cancel) directly for management decision-making.

** Additional repositories, live demos, and video walkthroughs available on portfolio: [my-portfolio-portfolio-iota.vercel.app](#)*

EDUCATION

Bachelor of Commerce – Accounting (English Section)

Grade: Very Good

MILITARY STATUS

Permanently Exempted